



3.2

Data Approximation and Neville's Method

Neville's Method

Disadvantages: No polynomial, we'll find the value only (unlike Lagrange)

(max to be in an exam)

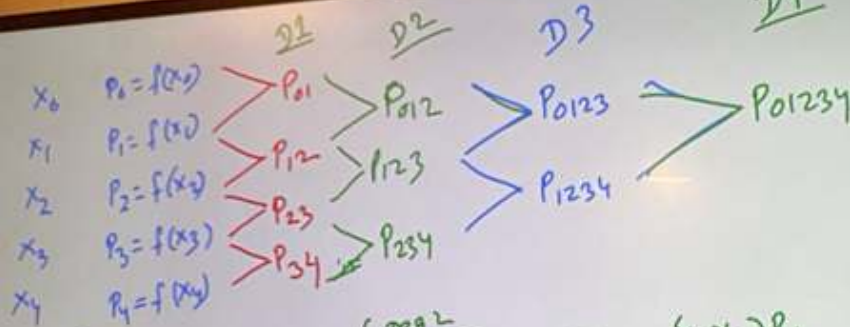
Degree 3⁰

$$P_{0123}(x) = \frac{(x-x_0)P_{123} - (x-x_3)P_{012}}{x_3 - x_0}$$

$$P_{1234}(x) = \frac{(x-x_0)P_{234} - (x-x_4)P_{123}}{x_4 - x_0}$$

Degree 4⁰

$$P_{01234}(x) = \frac{(x-x_0)P_{1234} - (x-x_4)P_{0123}}{x_4 - x_0}$$



Degree 1⁰

$$P_{01}(x) = \frac{(x-x_1)P_0 - (x-x_0)P_1}{x_1 - x_0}$$

$$P_{12}(x) = \frac{(x-x_1)P_2 - (x-x_2)P_1}{x_2 - x_1}$$

$$P_{23}(x) = \frac{(x-x_2)P_3 - (x-x_3)P_2}{x_3 - x_2}$$

$$P_{34}(x) = \frac{(x-x_3)P_4 - (x-x_4)P_3}{x_4 - x_3}$$

Degree 2⁰

$$P_{012}(x) = \frac{(x-x_2)P_{12} - (x-x_2)P_{01}}{x_2 - x_0}$$

$$P_{123}(x) = \frac{(x-x_1)P_{23} - (x-x_3)P_{12}}{x_3 - x_1}$$

$$P_{234}(x) = \frac{(x-x_2)P_{34} - (x-x_4)P_{23}}{x_4 - x_2}$$

x_0	$P_0 = Q_{0,0}$				
x_1	$P_1 = Q_{1,0}$	$P_{0,1} = Q_{1,1}$			
x_2	$P_2 = Q_{2,0}$	$P_{1,2} = Q_{2,1}$	$P_{0,1,2} = Q_{2,2}$		
x_3	$P_3 = Q_{3,0}$	$P_{2,3} = Q_{3,1}$	$P_{1,2,3} = Q_{3,2}$	$P_{0,1,2,3} = Q_{3,3}$	
x_4	$P_4 = Q_{4,0}$	$P_{3,4} = Q_{4,1}$	$P_{2,3,4} = Q_{4,2}$	$P_{1,2,3,4} = Q_{4,3}$	$P_{0,1,2,3,4} = Q_{4,4}$

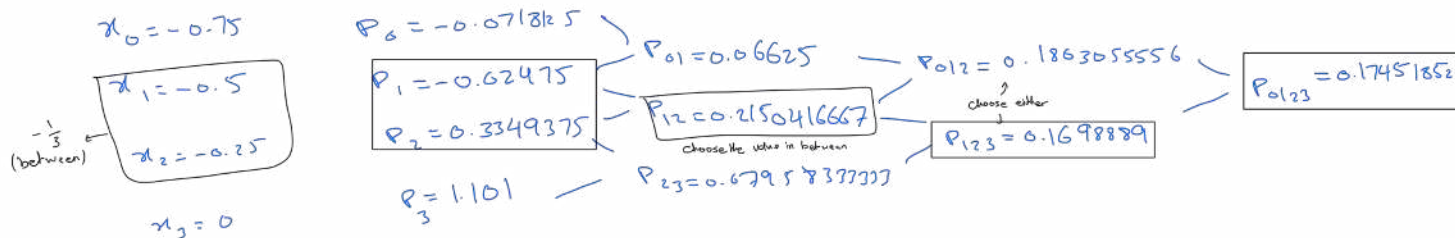
EXERCISE SET 3.2

1. Use Neville's method to obtain the approximations for Lagrange interpolating polynomials of degrees one, two, and three to approximate each of the following:
 - a. $f(8.4)$ if $f(8.1) = 16.94410$, $f(8.3) = 17.56492$, $f(8.6) = 18.50515$, $f(8.7) = 18.82091$
 - b. $f\left(-\frac{1}{3}\right)$ if $f(-0.75) = -0.07181250$, $f(-0.5) = -0.02475000$, $f(-0.25) = 0.33493750$, $f(0) = 1.10100000$
 - c. $f(0.25)$ if $f(0.1) = 0.62049958$, $f(0.2) = -0.28398668$, $f(0.3) = 0.00660095$, $f(0.4) = 0.24842440$
 - d. $f(0.9)$ if $f(0.6) = -0.17694460$, $f(0.7) = 0.01375227$, $f(0.8) = 0.22363362$, $f(1.0) = 0.65809197$

1. Use Neville's method to obtain the approximations for Lagrange interpolating polynomials of degrees one, two, and three to approximate each of the following:

b. $f(-\frac{1}{3})$ if $f(-0.75) = -0.07181250$, $f(-0.5) = -0.02475000$, $f(-0.25) = 0.33493750$, $f(0) = 1.10100000$

x_0 x_1 x_2 x_3
 -0.75 -0.5 -0.25 0
 P_0 P_1 P_2 P_3



$$P_{01}(-\frac{1}{3}) = \frac{(-\frac{1}{3} + 0.75)(-0.02475) - (-\frac{1}{3} + 0.5)(-0.0718125)}{-0.5 - (-0.75)} = 0.06625$$

$$P_{12}(-\frac{1}{3}) = \frac{(-\frac{1}{3} + 0.5)(0.3349375) - (-\frac{1}{3} + 0.25)(-0.02475)}{-0.25 + 0.5} = 0.2150416667$$

$$P_{23}(-\frac{1}{3}) = \frac{(-\frac{1}{3} + 0.25)(1.101) - (-\frac{1}{3} - 0)(0.3349375)}{0 + 0.25} = 0.0795833333$$

$$\frac{\text{Deg } 28}{P_{012}} = \frac{(-\frac{1}{3} + 0.75)(0.2150416667) - (-\frac{1}{3} + 0.25)(0.606625)}{-0.25 + 0.75} = 0.180305556$$

$$P_{123} = \frac{(-\frac{1}{3} + 0.5)(0.8795833333) - (-\frac{1}{3} - 0)(0.2150416667)}{0 + 0.5} = 0.1698889$$

$$\frac{\text{Deg } 30}{P_{0123}} = \frac{(-\frac{1}{3} + 0.75)(0.1698889) - (-\frac{1}{3} - 0)(0.180305556)}{0 + 0.75} = 0.17451852$$

3. Use Neville's method to approximate $\sqrt{3}$ with the following functions and values.
 - a. $f(x) = 3^x$ and the values $x_0 = -2$, $x_1 = -1$, $x_2 = 0$, $x_3 = 1$, and $x_4 = 2$.
 - b. $f(x) = \sqrt{x}$ and the values $x_0 = 0$, $x_1 = 1$, $x_2 = 2$, $x_3 = 4$, and $x_4 = 5$.
 - c. Compare the accuracy of the approximation in parts (a) and (b).
4. Let $P_3(x)$ be the interpolating polynomial for the data $(0, 0)$, $(0.5, y)$, $(1, 3)$, and $(2, 2)$. Find y if the coefficient of x^3 in $P_3(x)$ is 6.
5. Neville's method is used to approximate $f(0.4)$, giving the following table.

$x_0 = 0$	$P_0 = 1$				
$x_1 = 0.25$	$P_1 = 2$	$P_{01} = 2.6$			
$x_2 = 0.5$	P_2	$P_{1,2}$	$P_{0,1,2}$		
$x_3 = 0.75$	$P_3 = 8$	$P_{2,3} = 2.4$	$P_{1,2,3} = 2.96$	$P_{0,1,2,3} = 3.016$	

Determine $P_2 = f(0.5)$.

6. Neville's method is used to approximate $f(0.5)$, giving the following table.

$x_0 = 0$	$P_0 = 0$			
$x_1 = 0.4$	$P_1 = 2.8$	$P_{0,1} = 3.5$		
$x_2 = 0.7$	P_2	$P_{1,2}$	$P_{0,1,2} = \frac{27}{7}$	

Determine $P_2 = f(0.7)$.

7. Suppose $x_j = j$, for $j = 0, 1, 2, 3$, and it is known that

$$P_{0,1}(x) = 2x + 1, \quad P_{0,2}(x) = x + 1, \quad \text{and} \quad P_{1,2,3}(2.5) = 3.$$

Find $P_{0,1,2,3}(2.5)$.

3. Use Neville's method to approximate $\sqrt{3}$ with the following functions and values.

a. $f(x) = 3^x$ and the values $x_0 = -2$, $x_1 = -1$, $x_2 = 0$, $x_3 = 1$, and $x_4 = 2$.

$x_0 = -2$ $P_0 = 3^{-2} = \frac{1}{9}$
 $x_1 = -1$ $P_1 = 3^{-1} = \frac{1}{3}$
 $x_2 = 0$ $P_2 = 3^0 = 1$
 $x_3 = 1$ $P_3 = 3^1 = 3$
 $x_4 = 2$ $P_4 = 3^2 = 9$

$P_{01} = \frac{2}{3}$
 $P_{012} = \frac{3}{2}$
 $P_{12} = \frac{4}{3}$
 P_{23}
 P_{34}

$(x - x_0) P$

$$P_{01} = \frac{(\frac{1}{2} + 2) \frac{1}{3} - (\frac{1}{2} + 1) \frac{1}{9}}{-1 + 2} = \frac{2}{3}$$

$$P_{012} = \frac{(\frac{1}{2} + 2) \frac{4}{3} - (\frac{1}{2} - 0) \frac{2}{3}}{0 + 2} = \frac{3}{2}$$

$$P_{12} = \frac{(\frac{1}{2} + 1)(1) - (\frac{1}{2} - 0) \frac{1}{3}}{0 + 1} = \frac{4}{3}$$

b. $f(x) = \sqrt{x}$ and the values $x_0 = 0$, $x_1 = 1$, $x_2 = 2$, $x_3 = 4$, and $x_4 = 5$.

$$f(x) = \sqrt{x} = \sqrt{3}$$

$$x = 3$$

$$\begin{array}{ll} x_0 = 0 & p_0 = \sqrt{0} = 0 \\ x_1 = 1 & p_1 = \sqrt{1} = 1 \end{array} \quad \begin{array}{l} > \\ > \end{array} \quad \begin{array}{l} p_{0,1} = 3 \\ p_{0,1} = 3 \end{array}$$
$$\begin{array}{ll} x_2 = 2 & p_2 = \sqrt{2} \\ x_3 = 4 & p_3 = \sqrt{4} = 2 \\ x_4 = 5 & p_4 = \sqrt{5} \end{array}$$

$$p_{0,1} = \frac{(3-0)1 - (3-1)0}{1-0} = 3$$

5. Neville's method is used to approximate $f(0.4)$, giving the following table.

$x_0 = 0$	$P_0 = 1$			
$x_1 = 0.25$	$P_1 = 2$	$P_{01} = 2.6$		
$x_2 = 0.5$	$P_2 = 4$	$P_{1,2} = 3.2$	$P_{0,1,2} = 3.08$	
$x_3 = 0.75$	$P_3 = 8$	$P_{2,3} = 2.4$	$P_{1,2,3} = 2.96$	$P_{0,1,2,3} = 3.016$

$f(x_1)$



Start backwards to find P_2

Determine $P_2 = f(0.5)$.

$$P_{0123} = \frac{(x - x_0) P_{123} - (x - x_3) P_{012}}{x_3 - x_0}$$

$$3.016 = \frac{(0.4 - 0) 2.96 - (0.4 - 0.75) P_{012}}{0.75 - 0}$$

$P_{012} = 3.08$ using calculator

$$P_{012} = \frac{(x - x_0) P_{12} - (x - x_2) P_{01}}{x_2 - x_0}$$

$$3.08 = \frac{(0.4 - 0) P_{12} - (0.4 - 0.5) (2.6)}{0.5 - 0} \quad P_{12} = 3.2$$

$$P_{12} = \frac{(x - x_1) P_2 - (x - x_2) P_1}{x_2 - x_1}$$

$$3.2 = \frac{(0.4 - 0.25) P_2 - (0.4 - 0.5) (2)}{0.5 - 0.25} \quad P_2 = 4$$

7. Suppose $x_j = j$, for $j = 0, 1, 2, 3$, and it is known that

$$P_{0,1}(x) = 2x + 1, \quad P_{0,2}(x) = x + 1, \quad \text{and} \quad P_{1,2,3}(2.5) = 3.$$

Find $P_{0,1,2,3}(2.5)$

$x = 2.5$

$$P_{0,1}(2.5) = 2(2.5) + 1 = 6$$

$$P_{0,2}(2.5) = 2.5 + 1 = 3.5$$

$$P_{0,1,2}(2.5) = \frac{(x-x_0)P_{1,2} - (x-x_2)P_{0,1}}{x_1-x_0} = \frac{(2.5-0)3.5 - (2.5-2)(6)}{2-0} = \frac{8.75 - 3}{2} = 2.875$$

$$P_{0,1,2,3}(2.5) = \frac{(x-x_0)P_{1,2,3} - (x-x_3)P_{0,1,2}}{x_2-x_0} = \frac{(2.5-0)3 - (2.5-3)(2.875)}{3-0} = \frac{7.5 - (-0.875)}{3} = \frac{8.375}{3} = 2.791\bar{6}$$

⑦ $x_j = j, j = 0, 1, 2, 3$

$$P_{0,1}(x) = 2x + 1, \quad P_{0,2}(x) = x + 1, \quad P_{1,2,3}(2.5) = 3, \quad P_{0,1,2,3}(2.5) = ?$$

$$P_{0,1}(2.5) = 2(2.5) + 1 = 6$$

$$P_{0,2}(2.5) = 2.5 + 1 = 3.5$$

$$P_{1,2,3}(2.5) = 3$$

$$P_{0,1,2}(2.5) = \frac{(x-x_0)P_{1,2} - (x-x_2)P_{0,1}}{x_1-x_0} = \frac{(2.5-0)3.5 - (2.5-2)6}{2-0} = \frac{8.75 - 3}{2} = 2.875$$

$$P_{0,1,2,3}(2.5) = \frac{(x-x_0)P_{1,2,3} - (x-x_3)P_{0,1,2}}{x_2-x_0} = \frac{(2.5-0)3 - (2.5-3)2.875}{3-0} = \frac{7.5 - (-0.875)}{3} = \frac{8.375}{3} = 2.791\bar{6}$$